

QUBO Formulation: Brain-Inspired Binary Sparse Coding Problem

Brain-Inspired Binary Sparse Coding Problem

Problem Statement. Given an input pattern vector $y \in \mathbb{R}^M$ and an overcomplete dictionary of basis patterns $D = \{d_0, d_1, \dots, d_{N-1}\}$ where each $d_i \in \mathbb{R}^M$, the goal is to select a sparse binary subset of dictionary atoms that reconstructs the input pattern with minimal error. The reconstruction is formed by a linear combination of the selected atoms. The objective is to minimize the sum of the squared reconstruction error and a sparsity penalty proportional to the number of selected atoms.

Decision Variables. For each dictionary atom $i \in \{0, 1, \dots, N-1\}$, we introduce a binary decision variable $x_i \in \{0, 1\}$. The variable $x_i = 1$ indicates that the i -th dictionary atom d_i is selected for the reconstruction, and $x_i = 0$ indicates it is not selected.

QUBO Derivation. We derive the Quadratic Unconstrained Binary Optimization (QUBO) formulation by expanding the objective function and collecting terms.

Step 1: Original Objective. The objective is to minimize the reconstruction error plus a sparsity penalty:

$$\min_x \left\| y - \sum_{i=0}^{N-1} x_i d_i \right\|^2 + \lambda \sum_{i=0}^{N-1} x_i, \quad (1)$$

where $\lambda > 0$ is the sparsity regularization parameter. Expanding the squared norm yields:

$$\left\| y - \sum_{i=0}^{N-1} x_i d_i \right\|^2 = \left(y - \sum_{i=0}^{N-1} x_i d_i \right)^T \left(y - \sum_{j=0}^{N-1} x_j d_j \right) \quad (2)$$

$$= y^T y - 2y^T \sum_{i=0}^{N-1} x_i d_i + \left(\sum_{i=0}^{N-1} x_i d_i \right)^T \left(\sum_{j=0}^{N-1} x_j d_j \right) \quad (3)$$

$$= \|y\|^2 - 2 \sum_{i=0}^{N-1} (y^T d_i) x_i + \sum_{i=0}^{N-1} \sum_{j=0}^{N-1} (d_i^T d_j) x_i x_j. \quad (4)$$

Since $\|y\|^2$ is constant with respect to x , it can be omitted from the optimization. The objective becomes:

$$H(x) = \sum_{i=0}^{N-1} \sum_{j=0}^{N-1} (d_i^T d_j) x_i x_j - 2 \sum_{i=0}^{N-1} (y^T d_i) x_i + \lambda \sum_{i=0}^{N-1} x_i. \quad (5)$$

Step 2: Constraints. The problem is unconstrained; there are no constraints on the selection of atoms. Thus, no penalty terms are required.

Step 3: Penalty Terms. As there are no constraints, the penalty term is zero.

Step 4: Combined QUBO Hamiltonian. We combine the terms from Step 1. Using the idempotent property $x_i^2 = x_i$ for binary variables, we separate the quadratic and linear contributions. The double sum can be split into diagonal ($i = j$) and off-diagonal ($i \neq j$) terms:

$$H(x) = \sum_{i=0}^{N-1} (d_i^T d_i) x_i^2 + \sum_{i=0}^{N-1} \sum_{j \neq i}^{N-1} (d_i^T d_j) x_i x_j - 2 \sum_{i=0}^{N-1} (y^T d_i) x_i + \lambda \sum_{i=0}^{N-1} x_i \quad (6)$$

$$= \sum_{i=0}^{N-1} (d_i^T d_i) x_i + \sum_{i=0}^{N-1} \sum_{j \neq i}^{N-1} (d_i^T d_j) x_i x_j - 2 \sum_{i=0}^{N-1} (y^T d_i) x_i + \lambda \sum_{i=0}^{N-1} x_i. \quad (7)$$

Grouping the linear terms for each x_i :

$$H(x) = \sum_{i=0}^{N-1} [(d_i^T d_i) - 2(y^T d_i) + \lambda] x_i + \sum_{i=0}^{N-1} \sum_{j \neq i}^{N-1} (d_i^T d_j) x_i x_j. \quad (8)$$

To express this in standard QUBO form $H(x) = \sum_{i \leq j} Q_{i,j} x_i x_j + c$, we note that the off-diagonal sum counts each pair twice. We rewrite the off-diagonal sum as $2 \sum_{i < j} (d_i^T d_j) x_i x_j$. Thus:

$$H(x) = \sum_{i=0}^{N-1} [(d_i^T d_i) - 2(y^T d_i) + \lambda] x_i + \sum_{0 \leq i < j \leq N-1} 2(d_i^T d_j) x_i x_j. \quad (9)$$

Step 5: Q Matrix Entries. Reading off the coefficients from the expression above, the QUBO matrix entries $Q_{i,j}$ and offset c are given by:

$$Q_{i,i} = (d_i^T d_i) - 2(y^T d_i) + \lambda, \quad \text{for } i \in \{0, \dots, N-1\}, \quad (10)$$

$$Q_{i,j} = 2(d_i^T d_j), \quad \text{for } 0 \leq i < j \leq N-1, \quad (11)$$

$$c = 0. \quad (12)$$

Penalty Weight Justification. The formulation is unconstrained; therefore, no penalty weights are required to enforce constraints. The parameter λ serves as the sparsity regularization strength. It is chosen based on the desired trade-off between reconstruction fidelity and sparsity. A larger λ penalizes the selection of atoms more heavily, encouraging a sparser solution. In practice, λ is often tuned via cross-validation or set proportional to the noise level in the input pattern y . Since there are no constraints, λ does not need to satisfy a dominance condition over constraint violation penalties.

Correctness Argument. Minimizing the Hamiltonian $H(x)$ over the domain $\{0, 1\}^N$ recovers the optimal sparse coding solution because the QUBO formulation is an exact algebraic transformation of the original objective function. The quadratic terms $\sum_{i < j} 2(d_i^T d_j)x_i x_j$ correctly capture the interference between pairs of selected atoms in the reconstruction error, while the linear terms $(d_i^T d_i) - 2(y^T d_i) + \lambda$ account for the self-interaction of each atom, its alignment with the input, and the sparsity cost. Since the mapping is exact and unconstrained, the global minimum of $H(x)$ corresponds precisely to the binary vector x that minimizes the original reconstruction error plus sparsity penalty.

Illustrative Example. Consider a small instance with $N = 3$ dictionary atoms. The variables are $x_0, x_1, x_2 \in \{0, 1\}$. The QUBO matrix entries are instantiated from the general formulas as follows:

$$Q_{0,0} = (d_0^T d_0) - 2(y^T d_0) + \lambda, \quad (13)$$

$$Q_{1,1} = (d_1^T d_1) - 2(y^T d_1) + \lambda, \quad (14)$$

$$Q_{2,2} = (d_2^T d_2) - 2(y^T d_2) + \lambda, \quad (15)$$

$$Q_{0,1} = 2(d_0^T d_1), \quad (16)$$

$$Q_{0,2} = 2(d_0^T d_2), \quad (17)$$

$$Q_{1,2} = 2(d_1^T d_2). \quad (18)$$

The resulting Hamiltonian is:

$$H(x) = Q_{0,0}x_0 + Q_{1,1}x_1 + Q_{2,2}x_2 + 2Q_{0,1}x_0x_1 + 2Q_{0,2}x_0x_2 + 2Q_{1,2}x_1x_2. \quad (19)$$

Substituting the expressions for $Q_{i,j}$:

$$\begin{aligned} H(x) = & [(d_0^T d_0) - 2(y^T d_0) + \lambda] x_0 + [(d_1^T d_1) - 2(y^T d_1) + \lambda] x_1 + [(d_2^T d_2) - 2(y^T d_2) + \lambda] x_2 \\ & + 2(d_0^T d_1)x_0x_1 + 2(d_0^T d_2)x_0x_2 + 2(d_1^T d_2)x_1x_2. \end{aligned} \quad (20)$$

The offset is $c = 0$. The optimal solution x^* is the binary vector that minimizes this energy, yielding the selected dictionary atoms and the minimum sparse-coding energy.